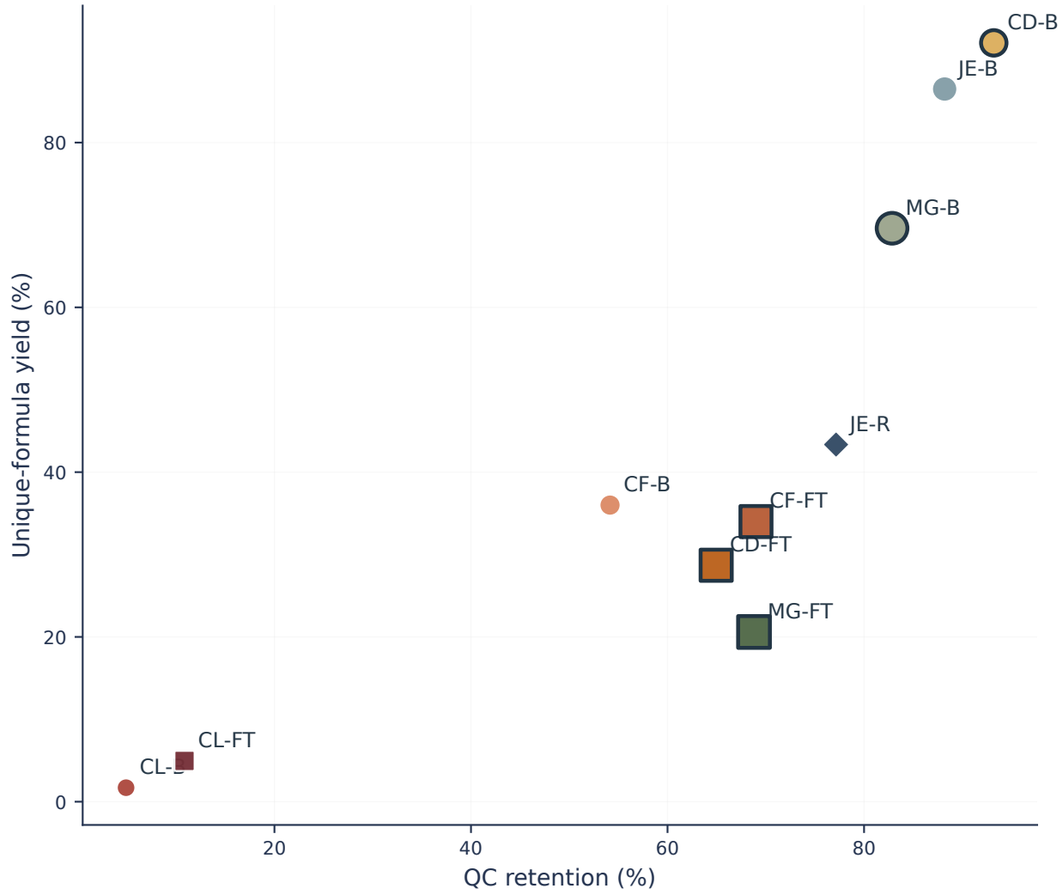
A Benchmark metrics

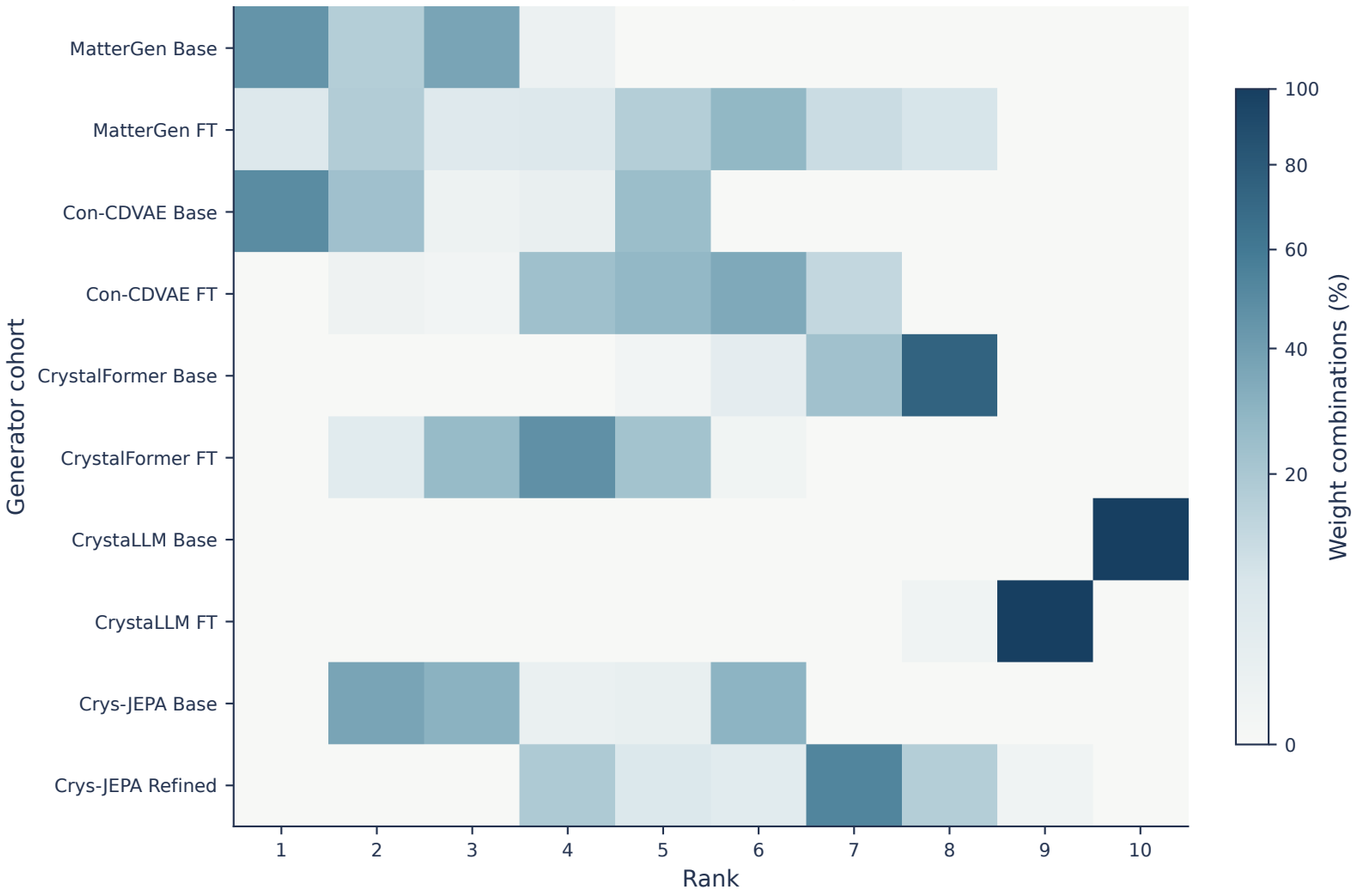
	QC retention	Unique formula yield	3d-containing yield
MatterGen Base	82.85%	69.60%	62.40%
MatterGen FT	68.80%	20.60%	68.70%
Con-CDVAE Base	93.20%	92.10%	34.30%
Con-CDVAE FT	64.95%	28.70%	64.95%
CrystalFormer Base	54.15%	36.00%	14.40%
CrystalFormer FT	69.00%	34.00%	64.55%
CrystaLLM Base	4.90%	1.70%	4.90%
CrystaLLM FT	10.85%	4.95%	10.80%
Crys-JEPA Base	88.20%	86.50%	32.65%
Crys-JEPA Refined	77.15%	43.35%	7.10%

Normalized score: 0.0, 0.2, 0.4, 0.6, 0.8, 1.0

B Structural yield vs diversity



C Rank stability



D First-place frequency

